

Exercise Solutions of Chp 2 N. Goldenfeld

Phase T. and RN

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1 Chapter 2

Exercise 2.1 Concerning the free energy of the n-n Ising model:

$$-H_{\Omega}(\{S\}) = H \sum_i S_i + J \sum_{i,j} S_i S_j. \quad (1.1)$$

(a) We show that $g(\beta) = \lim_{N(\Omega) \rightarrow \infty} \log(Z_{\Omega}(\beta))/N(\Omega)$ is convex down in β .

Recall

$$Z(\beta) = \sum_{\{S_i\}} e^{-\beta H_{\Omega}(\{S\})} = \sum_{\{S_i\}} e^{\beta H \sum_i S_i + \beta J \sum_{i,j} S_i S_j}. \quad (1.2)$$

Proceeding dead ahead, with the definition of convex down in mind, let $\beta_2 > \beta_1$ and $a, b \in [0, 1]$ such that $a + b = 1$ and consider

$$Z_{\Omega}(a\beta_1 + b\beta_2) = \sum_{\{S_i\}} [e^{(\beta_1 H \sum_i S_i + \beta_1 J \sum_{i,j} S_i S_j)^a} e^{(\beta_2 H \sum_i S_i + \beta_2 J \sum_{i,j} S_i S_j)^b}] \quad (1.3)$$

Now by Holder's inequality this can be bounded above as

$$\begin{aligned} \sum_{\{S_i\}} [e^{(\beta_1 H \sum_i S_i + \beta_1 J \sum_{i,j} S_i S_j)^a} e^{(\beta_2 H \sum_i S_i + \beta_2 J \sum_{i,j} S_i S_j)^b}] &\leq \left(\sum_{\{S_i\}} e^{(\beta_1 H \sum_i S_i + \beta_1 J \sum_{i,j} S_i S_j)} \right)^a \\ &\quad \times \left(\sum_{\{S_i\}} e^{(\beta_2 H \sum_i S_i + \beta_2 J \sum_{i,j} S_i S_j)} \right)^b \end{aligned} \quad (1.4)$$

But this is precisely the same as

$$Z_{\Omega}(a\beta_1 + b\beta_2) \leq Z_{\Omega}(\beta_1)^a Z_{\Omega}(\beta_2)^b. \quad (1.5)$$

It follows that

$$\log(Z_\Omega(a\beta_1 + b\beta_2)) \leq a \log(Z_\Omega(\beta_1)) + b \log(Z_\Omega(\beta_2)) \quad (1.6)$$

We then see immediately that

$$\begin{aligned} g(a\beta_1 + b\beta_2) &= \lim_{N(\Omega) \rightarrow \infty} \log(Z_\Omega(a\beta_1 + b\beta_2))/N(\Omega) \\ &\leq \lim_{N(\Omega) \rightarrow \infty} (a \log(Z_\Omega(a\beta_1))/N(\Omega) + b \log(Z_\Omega(b\beta_2))/N(\Omega)) \\ &= ag(\beta_1) + bg(\beta_2). \end{aligned}$$

and we are done.

(b) We will show that the free energy per unit volume

$$f(T) = -\frac{1}{\beta}g(\beta) = -kTg(1/(kT)) \quad (1.7)$$

is concave up in T .

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(c) We have the Gibbs free energy defined as

$$\Gamma_\Omega(M) = F_\Omega(H(M)) + N(\Omega)MH(M) \quad (1.8)$$

where $H(M)$ is defined implicitly as by $M(H) = -\partial f/\partial H$. We will show that $\Gamma(M)$ is convex down in M .